A Morphologically-Aware Transformer Network for High-Fidelity V1-V5 ECG Lead Generation

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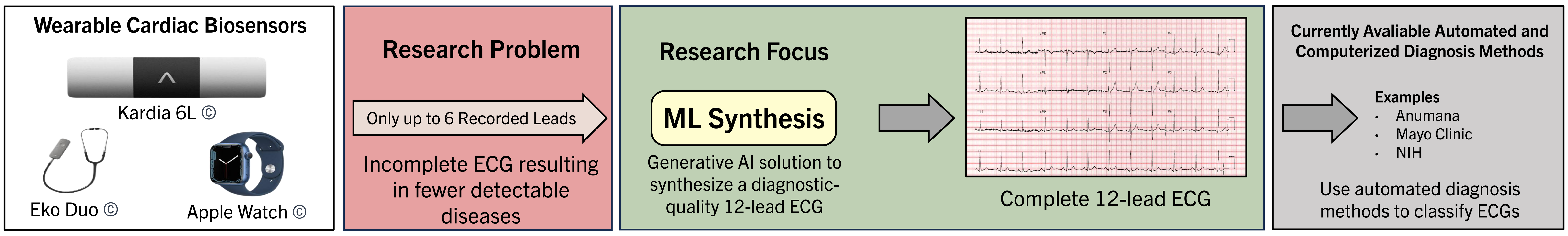
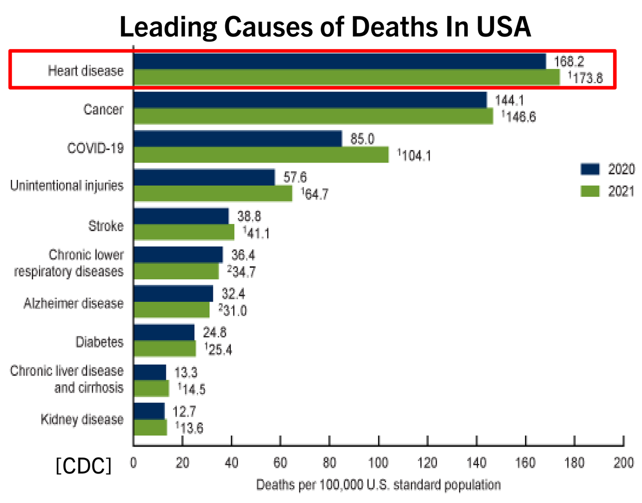
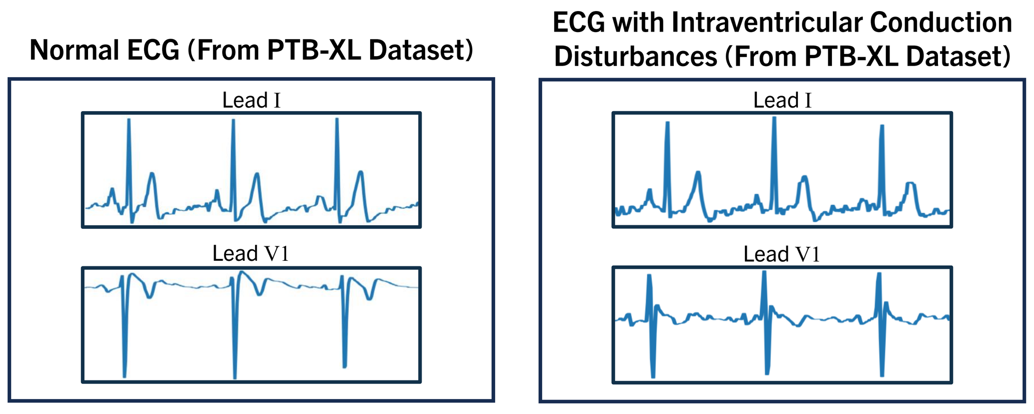
RESEARCH FOCUS



Cardiovascular disease (CVD) is the leading cause of death in the United States. Early detection and at home monitoring can prevent up to 80% of associated fatalities. Smart wearable devices are ideal for at-home monitoring but only record three leads, limiting detectable diseases. This project aims to create a generative AI solution to synthesize a diagnostic-quality 12-lead ECG, necessary for comprehensive CVD detection.

BACKGROUND



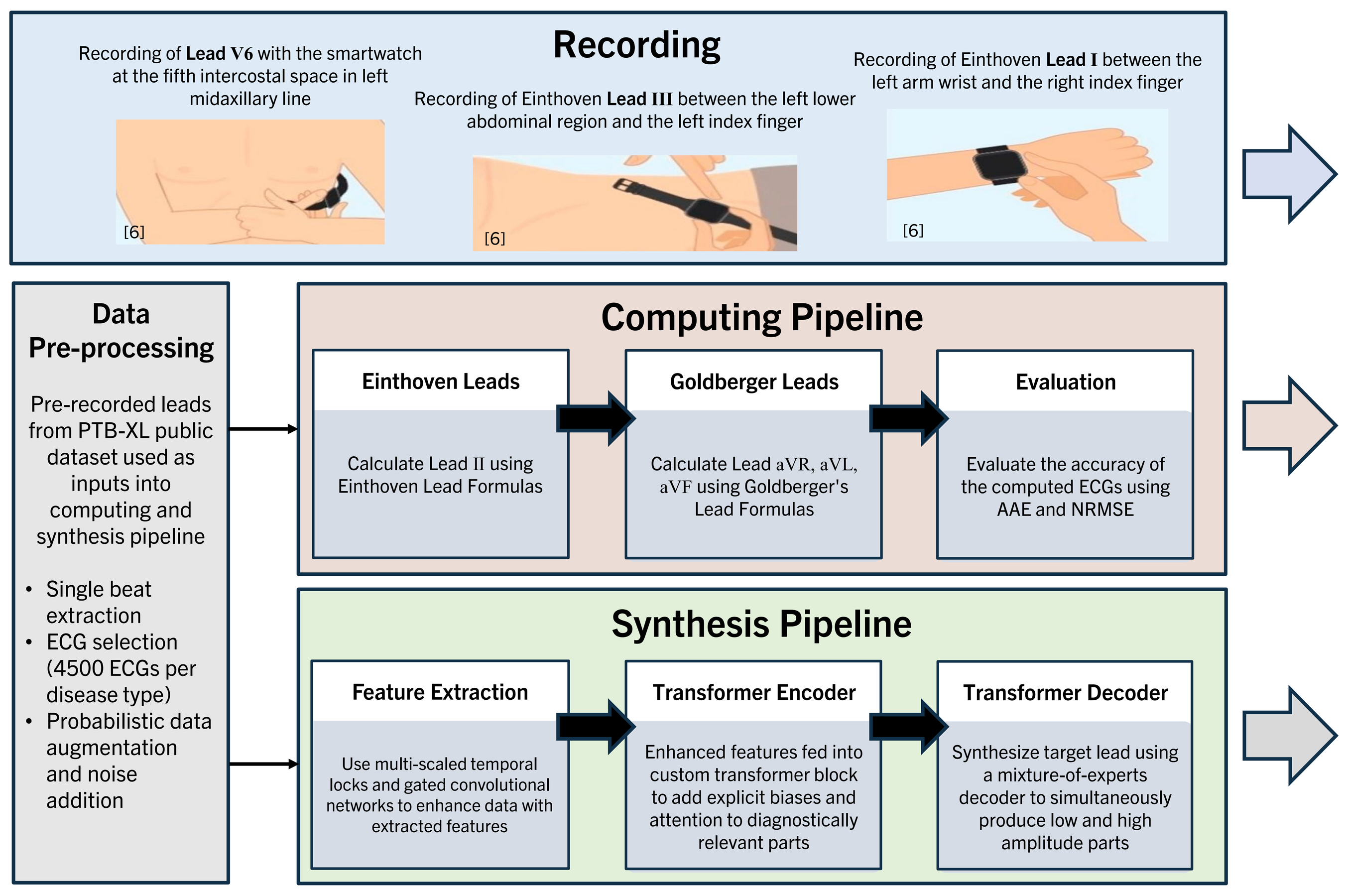


Example of Cardiovascular Disease that only shows up in Lead V1 while not showing up in Lead I, **highlighting the need for a full 12-lead ECG.** Plotted ECGs from the PTB-XL Dataset

Cardiovascular Disease (CVD) is the leading cause of death in the USA

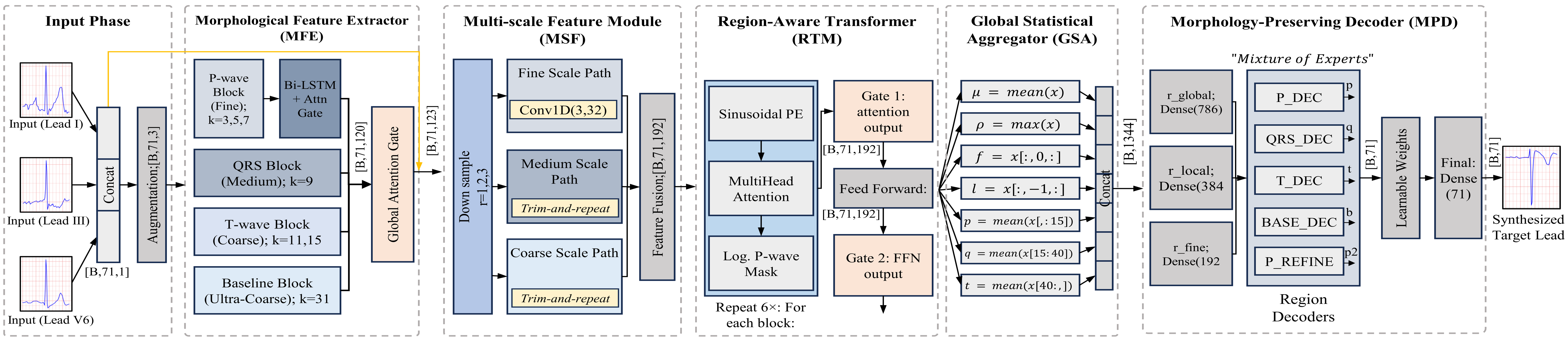
- At home monitoring is crucial for detecting life threatening diseases like Sudden Cardiac Arrest
  - Cardiovascular Disease (CVD) can show up only weeks in advance
- Many types of CVD like Ventricular diseases, pericarditis and types of Myocardial Infarction CANNOT be detected
- Asymptomatic individuals with CVD rarely get a full 12-lead ECG taken, even at annual check-ups
- Access to 12-Lead ECGs in low and middle-income countries could be as low as 10%

12-LEAD ECG GENERATION STRATEGY



Lead	Synthesis Method
I	Record
III	Record
V6	Record
II	Einthoven Lead System
aVR	Goldberger's Lead System
aVL	Goldberger's Lead System
aVF	Goldberger's Lead System
V1	Synthesize Using CardioNet
V2	Synthesize Using CardioNet
V3	Synthesize Using CardioNet
V4	Synthesize Using CardioNet
V5	Synthesize Using CardioNet

MORPHOLOGICALLY-AWARE TRANSFORMER NETWORK



(1) Morphological Feature Extractor (MFE) uses multi-resolution convolutions with dual saliency gates to capture global, local and P-wave ECG features. Designed to extract features from the ECG signals at multiple temporal resolutions using simultaneous convolutional streams, each utilizing distinct kernel widths which are adaptively fused and gated together

(2) Multi-scale Feature Module (MSF) aggregates features across fine, medium and course temporal scales. Transforms MFE output into fine, medium, and coarse-grained streams, each processed using with different strides and kernel sizes utilizing a "trim-and-repeat" alignment to minimize antialiasing

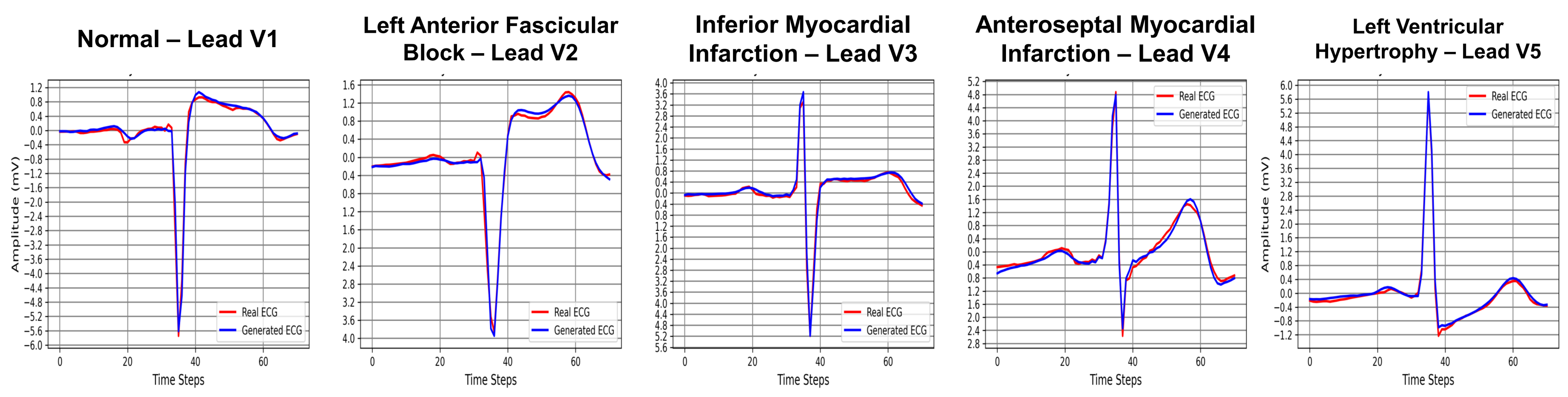
(3) Region-aware Transformer Module (RTM) utilizes muti-resolution convolutional and recurrent layers to extract features of from each ECG region. Processes multiscale tensor output of MSF using customized self-attention mechanism that uses explicit biases toward specific ECG phases

(4) Global Statistical Aggregator (GSA) distils the temporal feature maps into concise descriptors for each region. Integrates temporal sequences into a holistic feature vector by stacking seven complementary statistical descriptors

(5) Morphology Preserving Decoder (MPD) consists of a sub-decoder network to synthesize the distinct segments of the waveform of the target lead. Synthesizes the target lead using a mixture-of-experts decoder with five parallel linear experts, each specializing in different waveform families, which are combined softly via a learned gating matrix

(6) Loss Function & Training - Custom loss function blends various types penalties to accurately represent the ECG signals. Trained with Adam optimizer with global gradient norm clipping set to 1.0 and ReduceLROnPlateau scheduler

RESULTS



Leads	AAE of Synthesized Leads V1-V5					
	Normal	ASMI	LAFB	LVH	IMI	Average
Lead V1	0.0535	0.0548	0.0684	0.0427	0.0592	0.0557
Lead V2	0.0600	0.0749	0.0676	0.0619	0.0648	0.0659
Lead V3	0.0562	0.0862	0.0547	0.0604	0.0595	0.0634
Lead V4	0.0320	0.0733	0.0403	0.0399	0.0435	0.0458
Lead V5	0.0216	0.0455	0.0304	0.0220	0.0265	0.0292

Leads	NRMSE of Synthesized Leads V1-V5					
	Normal	ASMI	LAFB	LVH	IMI	Average
Lead V1	0.0535	0.0548	0.0684	0.0427	0.0592	0.0557
Lead V2	0.0600	0.0749	0.0676	0.0619	0.0648	0.0659
Lead V3	0.0562	0.0862	0.0547	0.0604	0.0595	0.0634
Lead V4	0.0320	0.0733	0.0403	0.0399	0.0435	0.0458
Lead V5	0.0216	0.0455	0.0304	0.0220	0.0265	0.0292

CONCLUSIONS

- The proposed model was designed to focus on the clinically significant features of an ECG with the goal of making the synthesized ECG leads of diagnostic quality
- This approach is able to synthesize ECGs that are morphologically similar, with low error rates, indicating that the model is able to capture the subtle variations that are clinically significant across all the subsegments of the ECG (ie. P, QRS, T)
- The use of a multi-stage architecture and a learned multi-component loss function was shown to be an effective method to capture heterogenous attributes associated with both morphology and timing, making it suitable for both automated diagnosis and ECG synthesis